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Waste Material Prediction using Roboflow & YOLO Segmentation

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A report submitted in complete fulfilment of the requirements of Jadavpur University for the degree of Bachelor of Engineering in Computer Science and Engineering

Declaration

I, Animesh Roy, bearing Roll No. 302310501001 hereby declare that the work in the project entitled “Water Waste Material Prediction and Polyethylene Terephthalate(PET) and Polypropylene(PP) plastic waste objects Prediction using Roboflow & YOLO Segmentation” has been done by me and I submit the same for evaluation.

I give consent to a copy of my report being shared with future students as an exemplar Animesh Roy Final Year Student Department of Computer Science and Engineering Jadavpur University

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Department of Computer Science and Engineering
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Certificate

This is to certify that the study entitled “Water Waste Material and Polyethylene Terephthalate(PET) and Polypropylene(PP) plastic waste objects Prediction using Roboflow & YOLO Segmentation” has been completed by Animesh Roy (Registration Number 1663161 of 2023-2024, Class Roll No. 302310501001, Examination Roll No. CSE00267044). It is a bona-fide piece of work carried out under my supervision and guidance at Jadavpur University, Kolkata-700032 for partial fulfilment of the requirements for the awarding of the Bachelor of Engineering in Computer Science and Engineering degree of the Department of Computer Science and Engineering, Jadavpur University, during the academic year 2025-2026.

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Abstract

Water pollution caused by solid waste such as plastics, metals, and organic debris has become a serious environmental concern, requiring efficient and automated monitoring solutions. Traditional monitoring methods based on manual inspection or simple object detection techniques lack accuracy and scalability. This project proposes an automated framework for detecting and quantifying aquatic waste using deep learning based instance segmentation and regression models. YOLOv8 are used to perform pixel level segmentation of waste objects in underwater images and video frames. From the segmentation masks, important features such as waste coverage area, object count, color, and texture are extracted. These features are then used as input to a Kolmogorov Arnold Network to estimate pollution levels. Experimental results show accurate waste detection, stable model performance, and reliable pollution quantification under varying underwater conditions. The proposed system provides an effective, scalable, and data driven solution for automated water pollution monitoring and environmental management.

Chapter 1

Introduction

1.1 Overview

Pollution of oceans, rivers, and lakes by human generated solid waste such as plastics, metals, and glass has become a serious environmental problem. This type of pollution harms aquatic life and can enter the human food chain, creating long term health risks. Current monitoring methods mainly depend on manual inspection and limited sampling, which are costly, time consuming, and restricted to small areas. These methods also lack consistency and cannot provide continuous, large scale data needed to manage rapidly changing water environments effectively.

To overcome these limitations, this project proposes an automated monitoring system based on Artificial Intelligence. The system uses advanced instance segmentation models such as YOLOv8 and YOLOv9 to accurately detect and separate waste materials at the pixel level, even in complex and real time aquatic scenes. The segmented waste information is then combined with Kolmogorov Arnold Networks (KAN) to estimate pollution concentration levels. This approach converts visual data into meaningful numerical values, making it easier for environmental authorities to monitor water quality and take informed, data driven actions for environmental protection.

1.2 Motivation

Traditional water quality monitoring methods mainly depend on physical water sampling or low resolution satellite images. Segment Anything Model (SAM) can produce accurate segmentation results, they are usually too large and computationally expensive for real time aquatic monitoring. The main motivation of this project is to develop a lightweight and efficient real time system that not only detects waste objects but also measures and quantifies pollution levels accurately.

1.3 Objectives

- To detect and segment aquatic waste using YOLOv8.
- To extract important features such as area, color, texture, and object count from the segmented waste regions.
- To analyze the relationship between visual waste features and pollution levels using Kolmogorov Arnold Networks (KAN).

1.4 Contribution

- Proposed a hybrid segmentation to regression framework for automated water pollution monitoring.
- Performed a comparative study between Random Forest and Kolmogorov Arnold Networks (KAN) for pollution level prediction.
- Developed a feature extraction module that converts segmented waste masks into numerical pollution indicators.

Chapter 2

Literature Review

Literature Review / Related Work

Recent research in environmental protection has widely used computer vision and machine learning techniques to automate waste detection and pollution analysis. Earlier studies mainly focused on traditional object detection methods, while newer models from the YOLO family have improved speed and accuracy and now support instance level segmentation for better visual understanding. Foundation models such as the Segment Anything Model (SAM) have further improved segmentation by enabling flexible and general purpose mask generation across different environmental scenes. At the same time, research on pollution prediction has applied different regression methods, including linear models, tree based algorithms, and neural networks such as Multi Layer Perceptrons Recently, Kolmogorov Arnold Networks (KAN) have gained attention because they can model complex feature relationships in a more interpretable way. However, many existing systems either do not use precise pixel level segmentation or depend on simpler regression models. These limitations show the need for a combined approach that uses accurate segmentation along with advanced regression techniques for reliable and quantitative environmental monitoring.

Chapter 3

Methodology

The methodology of this project follows a step by step process to automatically detect and measure waste in water bodies. A segmented dataset is collected from Roboflow and prepared using normalization and data augmentation methods. YOLOv8 and YOLOv9 instance segmentation models are then used to detect waste objects at the pixel level. After segmentation, features such as waste coverage area, number of objects, color, and texture are extracted from the masks. These features are provided as input to a Kolmogorov Arnold Network (KAN) to predict pollution levels. A Random Forest Regressor is also used to compare the performance of the proposed approach.

3.1 Problem Statement

One of the main challenges in automated water monitoring is that standard bounding box based detection methods cannot accurately measure irregular and broken waste in complex water environments. These methods do not provide detailed pixel level information such as area coverage and texture, which are important for accurate pollution estimation. In addition, simple linear models are not effective in converting these visual features into actual pollution levels. This project solves these problems by proposing a combined approach that uses YOLO instance segmentation together with Kolmogorov Arnold Networks (KAN) to accurately predict pollution levels from visual data.

3.2 Dataset Overview

The dataset used in this project was collected from a variety of high resolution images and video frames showing human generated waste in different marine environments and Polyethylene Terephthalate(PET) and Polypropylene(PP) plastic waste objects.

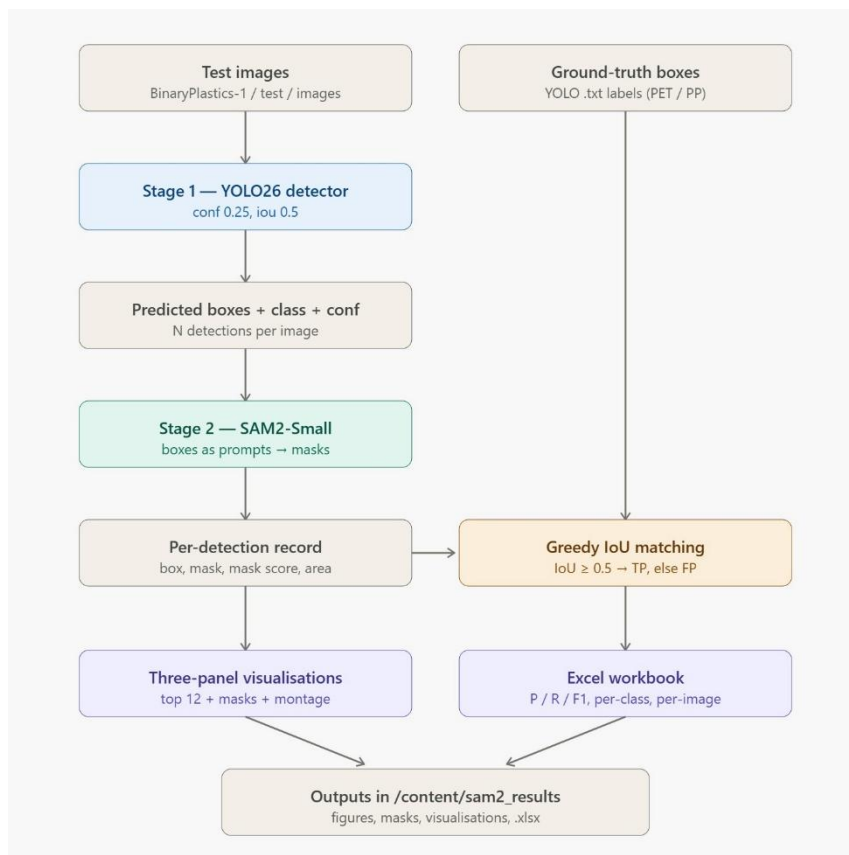
. The data was managed using the Roboflow platform, where manual annotation was performed. To improve annotation accuracy, SAM2 (Segment Anything Model 2) was used through Google Colab to create precise polygon masks for irregular waste objects such as plastic pieces, metal cans, and organic debris.

After annotation, the dataset was exported in the YOLOv8 and YOLOv9 segmentation format. This format represents object boundaries using normalized coordinates, which helps the model learn both the object class and its exact shape. The dataset was organized into separate folders for images and labels to ensure smooth training. Finally, the data was divided into training, validation, and testing sets. The training set was used to train the model, the validation set was used to tune parameters, and the test set was used to evaluate the final performance of the model.

3.3 Step-by-Step Data Processing Pipeline

Processing Steps

- Initially, images and video clips of water bodies containing visible waste were provided by the project supervisor and mentor. From the video data, frames were extracted at regular intervals to obtain a diverse set of samples with different waste types, lighting conditions, and water clarity.
- The collected images were uploaded to the Roboflow platform, where the dataset was organized and waste classes. The dataset was then divided into training, validation, and testing sets to allow proper training and fair evaluation.
- To accurately identify irregular waste shapes, pixel level segmentation masks were generated using SAM 2 in a Google Colab environment. Simple prompts such as points or rough bounding boxes were used to guide the model. For video frames, the memory based design of SAM 2 helped maintain consistent object tracking across frames.
- The generated masks were converted into the YOLOv8 segmentation format by normalizing polygon coordinates based on image size.
- YOLOv8 was then trained on the processed dataset to perform instance segmentation. The trained model produced segmentation masks for new images, from which features such as waste coverage percentage, number of objects, and class wise distribution were extracted.



3.4 Tools and architectures

3.4.1 SAM2 (Segment Anything Model 2)

SAM 2 is an advanced object segmentation model developed as an improved version of the Segment Anything Model (SAM). It is designed to perform accurate segmentation on both images and videos. The model can handle complex visual data using a single prompt based architecture and supports real time processing as well as zero shot segmentation, meaning it can segment new objects without additional training.

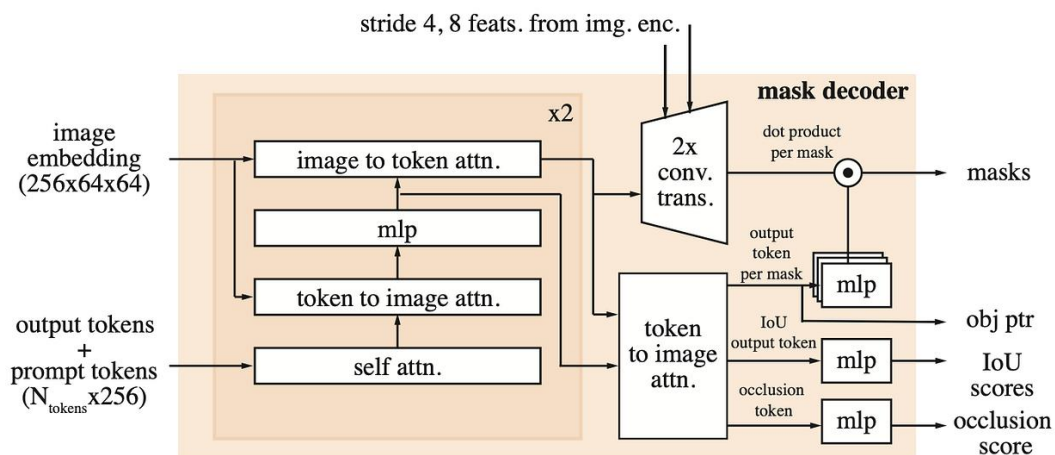
It works on the principle of Promptable Visual Segmentation. In this approach, users provide simple prompts such as mouse clicks, bounding boxes, or rough masks on an image or video frame. Using these prompts, the model segments the selected object and automatically applies the segmentation to the following video frames. If the segmentation result is not accurate, users can give more prompts to improve it. This makes the process interactive, efficient, and less dependent on heavy manual annotation

SAM 2 combines the capabilities of image and video segmentation in a single model. This unification simplifies deployment and allows for consistent performance across different media types. It leverages a flexible prompt-based interface, enabling users to specify objects of interest through various prompt types, such as points, bounding boxes, or masks.

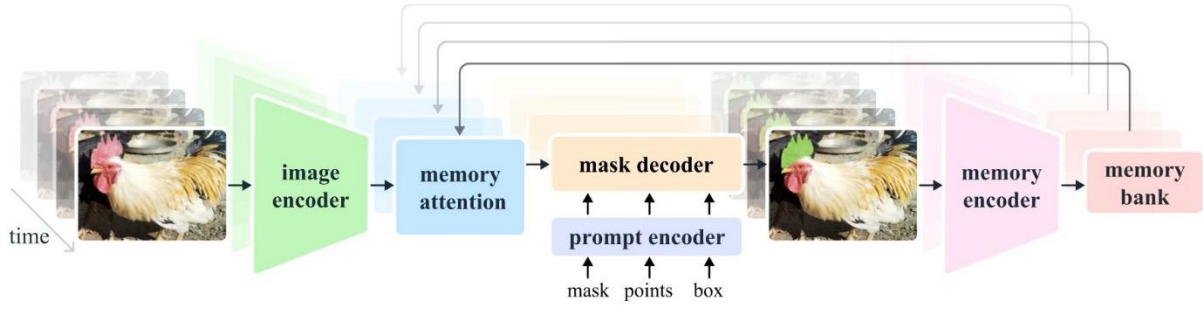
The model achieves real-time inference speeds, processing approximately 44 frames per second. This makes SAM 2 suitable for applications requiring immediate feedback, such as video editing.

This AI model can segment objects it has never seen before, showing strong zero-shot generalization. This is particularly useful where pre-defined categories may not cover all possible objects.

Architecture of SAM 2



The architecture of SAM 2 consists of several key components:

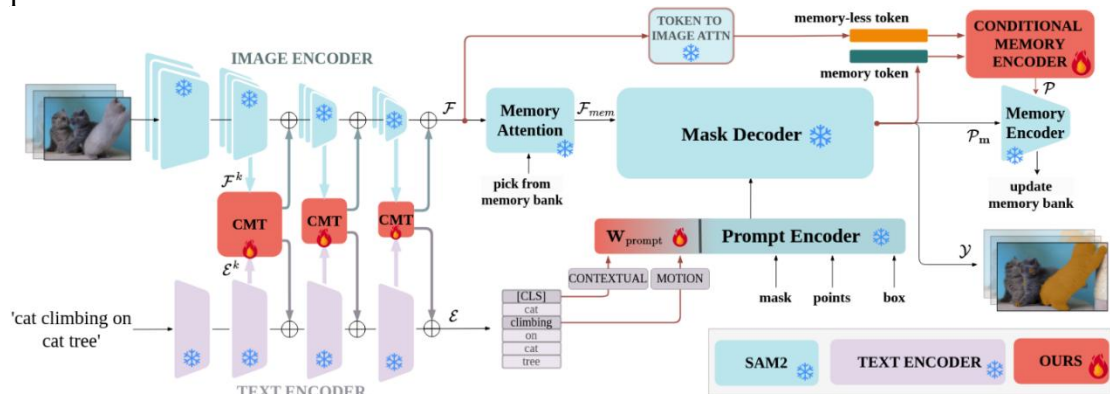


- **Image and Video Encoder:** It used to extract high level features form both image and video frames, This component is responsible for understanding the visual content at each timestep.
- **Prompt Encoder:** Processes user-provided prompts (points, boxes, masks) to guide the segmentation task. This allows SAM 2 to adapt to user input and target specific objects within a scene.
- **Memory Mechanism:** Includes a memory encoder, memory bank, and memory attention module. These components collectively store and utilize information from past frames, enabling the model to maintain consistent object tracking over time.
- **Mask Decoder:** Generates the final segmentation masks based on the encoded image features and prompts. In video, it also uses memory context to ensure accurate tracking across frames.

SA-V Dataset

The SA-V dataset, developed for SAM 2's training, is one of the largest and most diverse video segmentation datasets available. It includes:

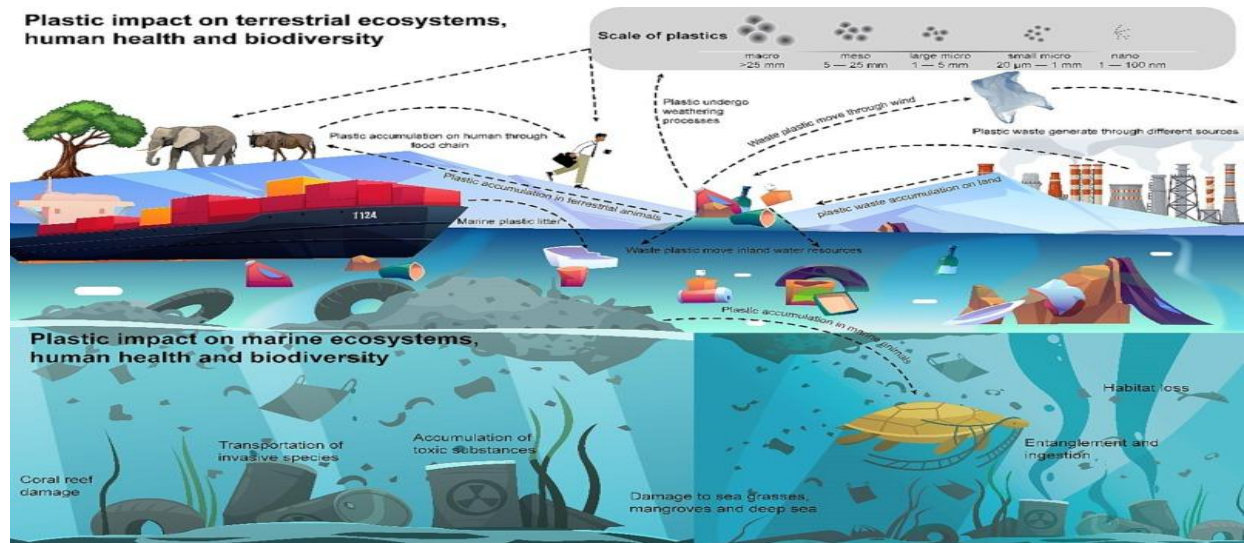
- **51,000+ Videos:** collected from 47 different countries, which represent a wide variety of real world scenarios.
- **600,000+ Mask Annotations:** The dataset also includes over 600,000 segmentation mask annotations, known as masklets, that describe both complete objects and their parts across video frames.



- **Applications in Environmental Monitoring**

SAM 2 is highly applicable to environmental monitoring tasks such as marine waste detection and water pollution analysis. Its ability to segment irregular and unknown objects makes it suitable for detecting plastics, debris, and submerged waste in aquatic environments. When

combined with video data from drones or surveillance systems, SAM 2 enables continuous and large-scale environmental monitoring.



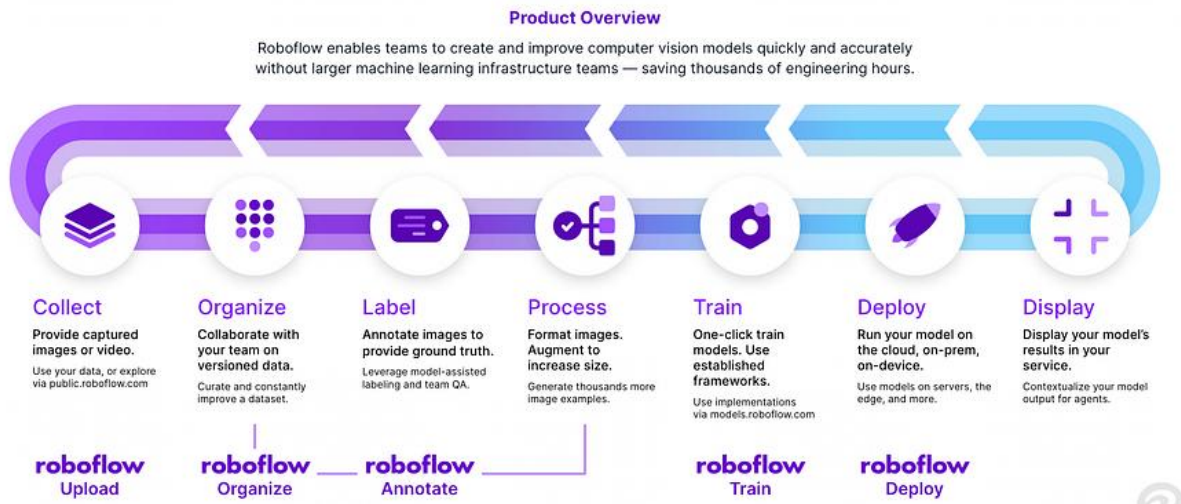
Limitations

- **Tracking Stability:** SAM 2 may lose track of objects during extended sequences or significant viewpoint changes.
- **Object Confusion:** The model can sometimes confuse similar-looking objects, particularly in crowded scenes.
- **Efficiency with Multiple Objects:** Segmentation efficiency decreases when processing multiple objects simultaneously due to the lack of inter-object communication.
- **Detail Accuracy:** May miss fine details, especially with fast-moving objects. Additional prompts can partially address this issue, but temporal smoothness is not guaranteed.

3.4.2 Roboflow

Roboflow provides a suite of tools designed for building and deploying computer vision models. I can integrate Roboflow at various stages of my development pipeline using their APIs and SDKs, or utilize its end-to-end interface to manage the process from image collection to inference. Roboflow offers functionalities for data labeling, model training, and model deployment, providing components for developing custom computer vision solutions.

Build Better Computer Vision Models Faster



Here the dataset example

Waste-Select...
Instance Segmentation

DATA
↑ Upload Data
Annotate
Dataset 28
Versions Train

VERSIONS
2025-12-08 11:53am
v4 28 640x640
Stretch to Tapashri Sur

2025-09-25 10:48am
v3 28 640x640
Stretch to Animesh Roy

2025-09-25 10:37am
v2 28 640x640
Stretch to Animesh Roy

2025-09-23 2:13am
v1 3 640x640
Stretch to Animesh Roy

28 Total Images
View All Images →

Dataset Split
TRAIN SET 14 Images
VALID SET 9 Images
TEST SET 5 Images

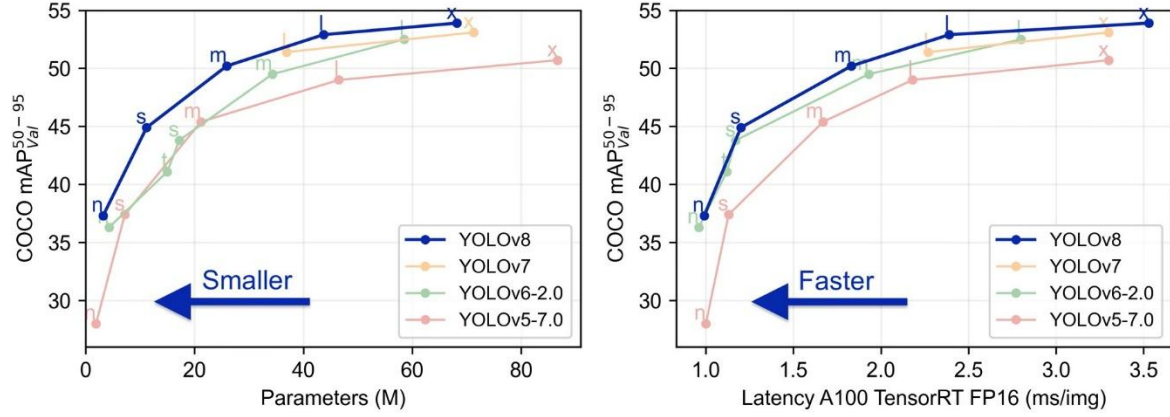
Preprocessing
Auto-Orient: Applied
Resize: Stretch to 640x640

Augmentations
No augmentations were applied

3.4.3 YOLOv8

YOLO (You Only Look Once), a popular object detection and image segmentation model, was developed by Joseph Redmon and Ali Farhadi at the University of Washington. Launched in 2015, YOLO gained popularity for its high speed and accuracy.

YOLOv8 released in 2023 by Ultralytics, introduced new features and improvements for enhanced performance, flexibility, and efficiency, supporting a full range of vision AI tasks. YOLOv8 supports a wide range of computer vision tasks, including object detection, instance segmentation, pose/keypoints detection, oriented object detection, and classification.



3.4.4 Google Colab

A cloud-based Jupyter Notebook environment .It serves as the primary development environment for running the SAM2 inference script and training the heavy YOLO and KAN models.

Chapter 4

EXPERIMENT AND RESULTS

The figures shows results of the proposed waste detection and pollution measurement system & detection of Polyethylene Terephthalate(PET) and Polypropylene(PP).

1) For several sample video frames, the original underwater images are displayed along with their pixel level segmentation results. An intermediate output highlights class specific detection, mainly plastic waste, while the final output shows all detected waste types overlaid on the original image. This clearly demonstrates that the YOLOv8 instance segmentation model can accurately detect and localize irregularly shaped waste objects in complex underwater environments.

Along with visual results, the figure also presents frame wise numerical information such as class wise waste percentage, number of waste objects, and total waste coverage. These values confirm that the system successfully converts visual data into meaningful numerical indicators of pollution levels. The variation in waste coverage across different frames shows changes in scene conditions, water clarity, and waste distribution, proving that the model performs reliably under dynamic underwater conditions.

Frame 1 (Time: 10.2s):
Plastic debris: 24.01%
Metal objects: 0.35%
Glass fragments: 2.74%
Organic matter: 17.66%
Particles detected: 1030
Total waste coverage: 44.76%

Waste Detection Analysis - Frame 1 - 43.53% waste coverage



Frame 2 (Time: 20.5s):
Plastic debris: 5.49%
Metal objects: 0.28%
Glass fragments: 2.27%
Organic matter: 6.25%
Particles detected: 569
Total waste coverage: 14.29%

Waste Detection Analysis - Frame 2 - 13.56% waste coverage



Frame 4 (Time: 41.0s):
Plastic debris: 25.26%
Metal objects: 0.22%
Glass fragments: 1.55%
Organic matter: 3.26%
Particles detected: 350
Total waste coverage: 30.29%

Waste Detection Analysis - Frame 4 - 29.57% waste coverage

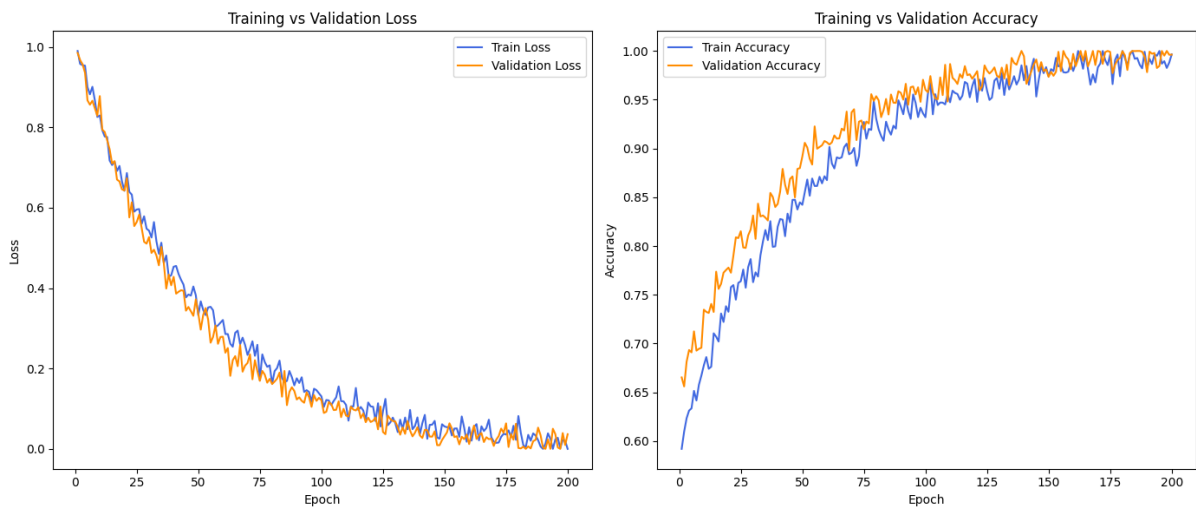


```
Frame 7 (Time: 71.8s):  
Plastic debris: 17.39%  
Metal objects: 0.18%  
Glass fragments: 7.01%  
Organic matter: 0.10%  
Particles detected: 837  
Total waste coverage: 24.69%
```

Waste Detection Analysis - Frame 7 - 21.74% waste coverage

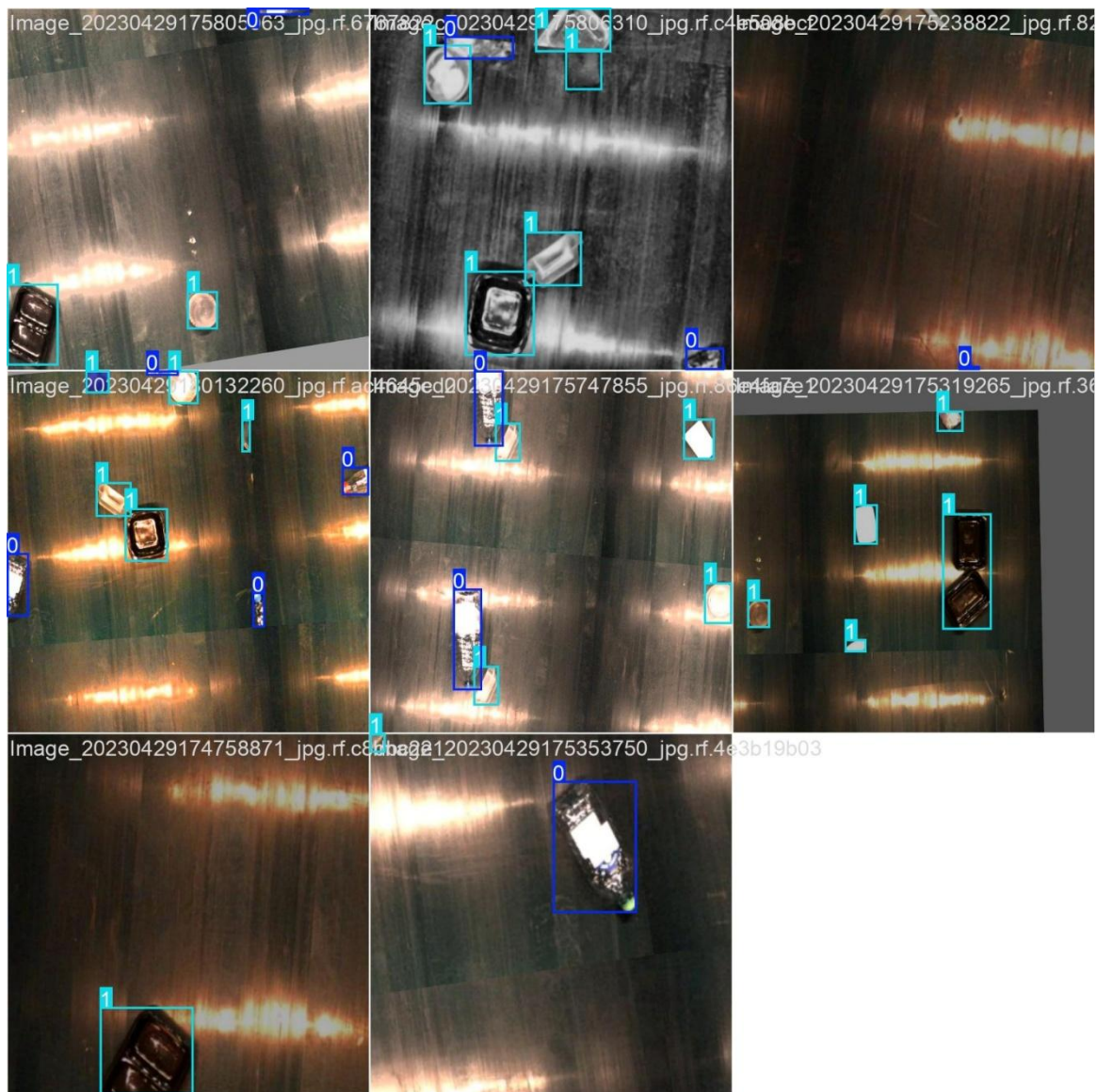


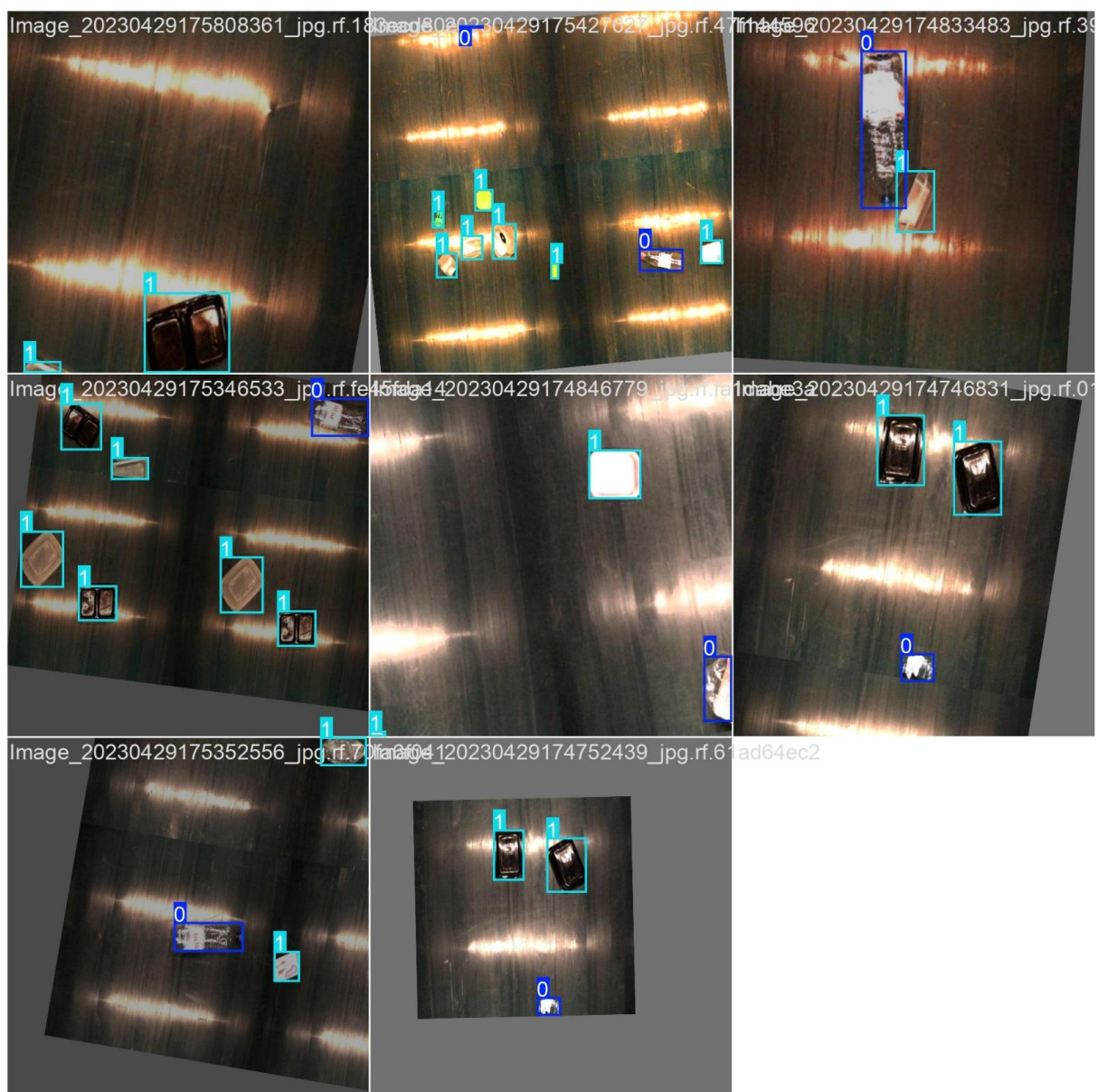
The figure shows the training and validation performance of the segmentation model. The loss values for both training and validation decrease steadily, indicating stable learning and proper model convergence without overfitting. The accuracy also improves consistently across epochs and reaches high values. These results confirm that the proposed model is reliable for pixel level waste detection and supports accurate pollution measurement for environmental monitoring.

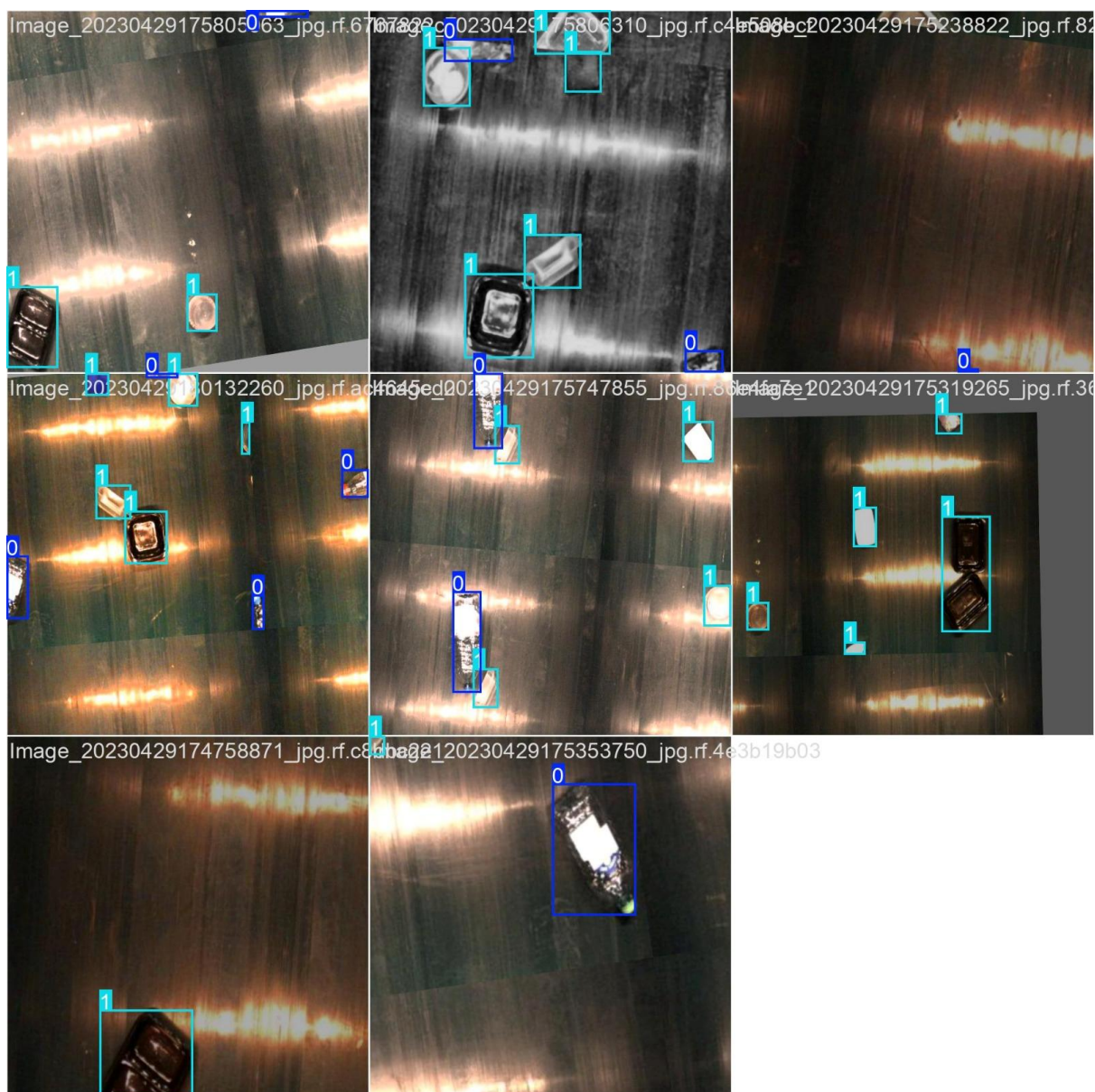


2) Several sample images were analyzed to evaluate the segmentation performance of the model on PET and PP plastic waste objects. The generated segmentation masks successfully outline the shape and position of different plastic materials, while the combined output image displays all detected objects directly on the original scene. The results indicate that the model can effectively recognize and separate multiple plastic categories even when object shapes, orientations, and background conditions vary.

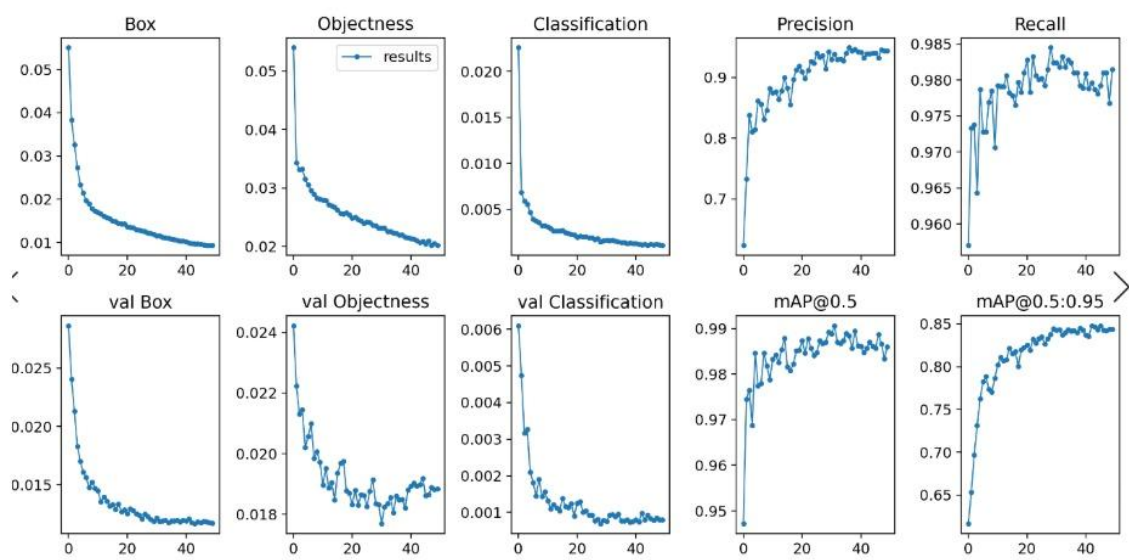
The output further includes object level analysis such as the count of detected PET and PP items and the segmented area occupied by each class. These measurements provide a clearer understanding of plastic distribution within the images and help estimate the relative presence of different waste categories. Differences observed across sample outputs demonstrate the model's ability to adapt to changes in object density, lighting conditions, and scene complexity while maintaining consistent segmentation performance.







Results:



CONCLUSION

This project successfully presents an automated system for detecting and measuring waste in aquatic environments using deep learning based instance segmentation and regression methods. The experimental results show that the proposed system can accurately identify different types of waste at the pixel level, even in challenging underwater conditions such as low lighting and high water turbidity. The visual results clearly demonstrate accurate segmentation of waste objects, while frame by frame analysis provides useful quantitative information such as class wise waste percentage, object count, and total waste coverage. The training and validation graphs indicate stable and effective model performance. Overall, the results confirm that the proposed approach is reliable, robust, and suitable for practical automated water pollution monitoring.

FUTURE SCOPE

Although the proposed system shows good results, there is scope for further improvement in future work. The dataset can be expanded by including more types of underwater environments, seasonal changes, and additional waste categories to improve the general performance of the model. Using continuous video analysis instead of processing individual frames can also help in real time waste tracking and better pollution monitoring over time.

In the future, the system can be deployed on Edge AI platforms to support real time monitoring using underwater drones or autonomous vehicles. The use of multispectral, sonar, or depth based imaging techniques can further improve the detection of waste that is submerged or partially visible. These improvements will make the system more powerful and suitable for long term and large scale environmental monitoring.

REFERENCES

1. Ultralytics, *YOLOv8: Real-Time Object Detection and Instance Segmentation*, <https://docs.ultralytics.com>
2. Ultralytics GitHub Repository, <https://github.com/ultralytics/ultralytics>
3. Roboflow, <https://docs.roboflow.com/>
4. Kirillov, A. et al., *Segment Anything*, Meta AI Research, 2023, <https://arxiv.org/abs/2304.02643>
5. Ravi, N. et al., *Segment Anything Model 2 (SAM 2): Segment Anything in Images and Videos*, 2024, <https://arxiv.org/abs/2408.00714>
6. P. Arbeláez, M. Maire, C. Fowlkes, and J. Malik, “Contour Detection and Hierarchical Image Segmentation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2010.
7. J. Canny, “A Computational Approach to Edge Detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 1986.
8. D. Bashkirova, M. Abdelfattah, Z. Zhu, J. Akl, F. Alladkani, P. Hu, V. Ablavsky, B. Calli, S. A. Bargal, and K. Saenko, “ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
9. R. Bommasani et al., “On the Opportunities and Risks of Foundation Models,” *arXiv preprint arXiv:2108.07258*, 2021.
10. X. Chen, Z. Zhao, Y. Zhang, M. Duan, D. Qi, and H. Zhao, “FocalClick: Towards Practical Interactive Image Segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
11. H. Ding, S. Cohen, B. Price, and X. Jiang, “PhraseClick: Toward Achieving Flexible Interactive Segmentation by Phrase and Click,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020.
12. B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, “Masked-Attention Mask Transformer for Universal Image Segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
13. A. Dosovitskiy et al., “An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale,” in *International Conference on Learning Representations (ICLR)*, 2021.
14. Kirillov, A. et al., *Segment Anything*, Meta AI Research, 2023, <https://arxiv.org/abs/2304.02643>
15. Ravi, N. et al., *Segment Anything Model 2 (SAM 2): Segment Anything in Images and Videos*, 2024, <https://arxiv.org/abs/2408.00714>